

Dyslexia, ADHD, Dyscalculia, and Dyspraxia Prediction in Children Using Machine Learning and Explainable AI

Debarati Dhar
Department of CSE
Ctg. Univ. of Eng. and Tec. (CUET)
Raouzan, Bangladesh
u1904038@student.cuet.ac.bd

Mahfuzulhoq Chowdhury
Department of CSE
Ctg. Univ. of Eng. and Tec. (CUET)
Raouzan, Bangladesh
mahfuz_csecuet@yahoo.com

Salman Farsi
Department of CSE
Ctg. Univ. of Eng. and Tec.
Raouzan, Bangladesh
u1804102@student.cuet.ac.bd

Adid Rahman
Department of CSE
Ctg. Univ. of Eng. and Tec. (CUET)
Raouzan, Bangladesh
u2004082@student.cuet.ac.bd

Abid Sarker Dipto
Department of CSE
Ctg. Univ. of Eng. and Tec. (CUET)
Raouzan, Bangladesh
u2004121@student.cuet.ac.bd

Suriya Sultana
Department of CSE
Ctg. Univ. of Eng. and Tec. (CUET)
Raouzan, Bangladesh
22mcse026@student.cuet.ac.bd

Abstract— Numerous neurodevelopment disorders, including dyslexia, ADHD, dyscalculia, and dyspraxia, impact a considerable proportion of children. The existing literary works on this disorder detection has a variety of limitations, including poor accuracy, insufficient feature selection, and class imbalance issue. To predict dyslexia, ADHD, dyscalculia, and dyspraxia, this paper proposed a ML based multi class prediction scheme using stacking ensemble approach. The work employed SMOTE to overcome class imbalance issue and RFE method for suitable feature selection. Hyper parameter optimization strategies are adapted and a number of ML algorithms were rigorously tested. With 88.22% precision and 89.35% accuracy, a stacking ensemble technique comprising Random Forest, XGBoost, LightGBM, ANN, and Logistic Regression produced the best prediction results. The proposed stacking ensemble technique provided at least 2% higher accuracy value than existing works. SHAP based model interpretation results also provided in this paper.

Keywords— Neurodevelopment disorders, Dyslexia, ADHD, Machine Learning, Feature engineering, Stacked Ensemble.

I. INTRODUCTION

Neurodevelopment disorders (NDDs), which affect millions of children worldwide and put a significant burden on individuals, families, and healthcare systems, are among the most significant global health concerns of the twenty-first century. According to the WHO, over 3 billion people worldwide suffered from neurodevelopment diseases in 2021. The common types of neurodevelopment disorder include dyslexia, ADHD, dyscalculia, and dyspraxia [1]. Approximately 10% of people worldwide suffer from dyslexia, the most common learning disability, whereas 7% of school-age children worldwide are thought to have ADHD, and 7 to 15% of the same population suffers from dysgraphia. Since learning impairments are frequently present in 30 to 50 percent of children with ADHD, the two disorders frequently co-occur, making diagnosis and treatment challenging for medical professionals [2]. Almost 290 million children and adolescents were suffered from aforementioned four neurodevelopment disorders in 2017, with 94% of them residing in South Asia and sub-Saharan

Africa [3]. Children with ADHD, dyslexia, dyspraxia, and dyscalculia have unique challenges, especially in Bangladesh. The number of ADHD and neurodevelopment disorder patients among Bangladeshi children's is more than thirty percent, according to [4]. Early detection and intervention are key to the effective management of neurodevelopment problems. Due to a shortage of qualified medical personnel, traditional methods for detecting neurodevelopment disorders are costly, time-consuming, and may not be culturally appropriate [5]. The remarkable ability of machine learning (ML) algorithms to uncover complex patterns in behavioral, cognitive, and neurodevelopment data makes them particularly well-suited for early detection and screening applications of ADHD, dyslexia, dyspraxia, and dyscalculia. There are a number of issues with the existing research that are currently available for predicting neurodevelopment disorders ([6], [7], [8]). Their primary focus was the classification of binary neurodevelopment disorders. Their disease prediction method's accuracy results are insufficient. Additionally, their work's feature sets and acquired data are small. The work in [9], [10], and [11] did not use any appropriate feature selection approach, explainable AI technique, hyper parameter tuning strategy, or class imbalance issue fixing technique. This paper provides a multi-class prediction method for numerous neurodevelopment disorders (Dyslexia, ADHD, Dyscalculia, and Dyspraxia) utilizing ML and explainable AI in order to overcome the current constraints. The following is a comprehensive summary of contributions: (i) This paper created a comprehensive, clinically validated dataset that includes extensive features that are useful for forecasting illnesses such as ADHD, dyslexia, dyscalculia, and dyspraxia.

(ii) This paper used Recursive Feature Elimination (RFE) technique for suitable feature selection. Class imbalance is addressed by using the SMOTE approach. Grid and Randomized Search are used for hyper parameter tuning.

(iii) This work examined a number of ML models and chooses the best one for multi-class neurodevelopment disease prediction.

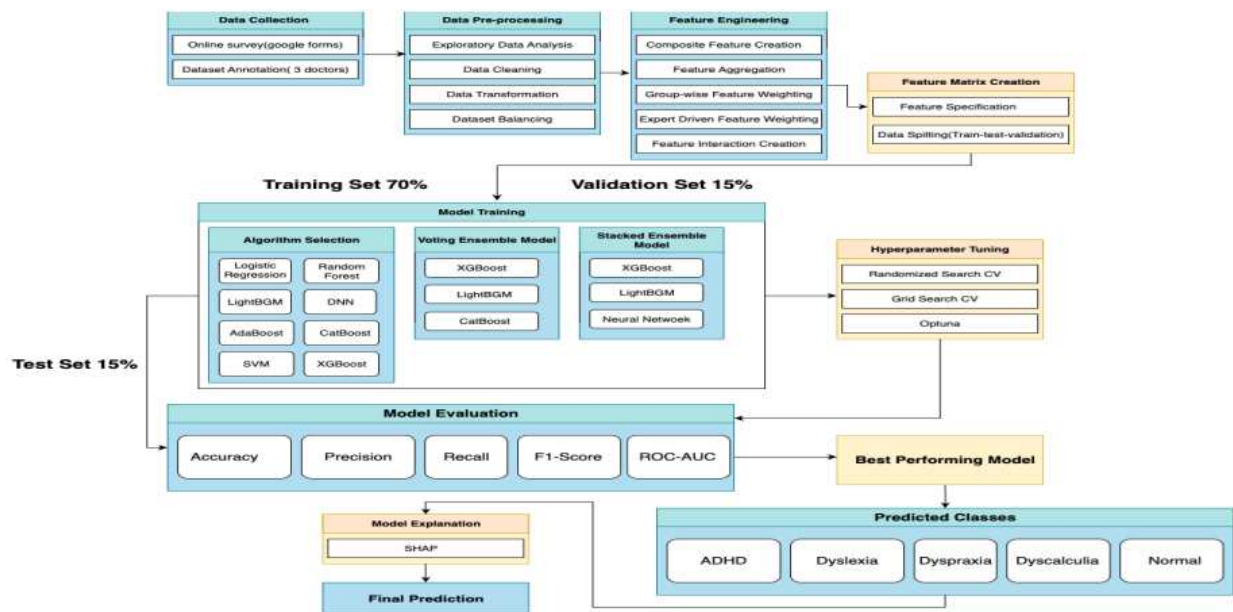


Fig. 1. Proposed neurodevelopment disorder prediction model

A stacking ensemble technique (Random Forest, XGBoost, LightGBM, ANN, and Logistic Regression) is employed in this paper since it outperformed other models. By clarifying the prediction models' decision-making procedures, the SHAP-based EAI model improves clinical interpretability. A general summary of the relevant work is provided in Section two. The proposed ML based prediction model operation is outlined in third section. The proposed system's performance is assessed in section 4. Final summary of the research are disclosed in Section five.

II. LITERATURE REVIEW

This section discusses the literary articles pertaining to neurodevelopment disorders. In order to diagnose ADHD, the study in [6] used NLP techniques. Nine ML algorithms for early ADHD diagnosis were compared in the article in [7]. The classification challenge was handled by the GBM method. Their model's accuracy is 85.29%. Using Reddit social media data, the study in [8] presented a novel artificial neural network (ANN) framework to predict adult ADHD. Among the models used to predict ADHD, the ANN had the highest accuracy (87.30 percent). Nevertheless, the study had other drawbacks, such as a small effective sample size, no external validation procedures and no feature-importance analysis. The research article [9] lists dyscalculia tools that are boosted by AI. They predicted dyscalculia using CNN and LSTM. Some shortcomings of this work include inconsistent data pretreatment pipelines, restricted feature exploration, inadequate domain-specific feature selection, and lack of interpretability techniques. An ANN was utilized in the work in [10] to diagnose dyslexia. The problem of dataset over fitting was not resolved by them. A multimodal machine learning pipeline was shown in [11] to predict neurodevelopment disorders, including dyslexia, dyscalculia, and ADHD. In their research, the logistic

regression model had the best accuracy (72.5 percent). Additionally, they didn't employ any explainable AI or clever feature selection techniques. The ensemble model had the best diagnosis accuracy in their work. A gradient boosting-based learning disability prediction method was introduced in [12]. Other neurodevelopment disorders were not addressed by them. They didn't use explainable AI, hyper parameter optimization, or appropriate feature selection. The work in [13] presented a dimensional-phenotypic framework that uses ensemble ML model in a multi-label setting to predict the risk of dyslexia and dyscalculia. They did not demonstrate the accuracy or F1 score of their suggested model. The lack of class imbalance problem solving and feature selection is a drawback of the works in [14], [15]. In order to predict ADHD from the EEG signal, the work in [16] employed CNN, Bi-LSTM, and PCA methods. A deep learning algorithm based on Yolo was created in [17] to diagnose ADHD in handwriting. Their suggested model's accuracy is a pitiful 72%. A model for predicting dyslexia severity based on random forests was provided in the article in [18]. They didn't work on predicting other neurodevelopment disorders. A CNN-based dyslexia prediction framework based on the EEG signal was given in the study in [19]. The paper in [20] employed a trained CNN model based on VGG 16 to detect dysgraphia in handwritten samples with a satisfactory accuracy level. The work in [21] used EEG signal for dyslexia and work in [22] developed web app for dyspraxia detection. The work in [23] investigated the impact of neurodevelopment disorder on software engineer. Predicting dyslexia, ADHD, dyscalculia, and dyspraxia disorders simultaneously was not addressed in the previous research. They didn't use any intelligent feature selection techniques. Using sophisticated feature selection, class imbalance solving, and hyper parameter tuning, this research proposes a ML and EAI based multi-class neurodevelopment disease prediction technique to address the current constraints.

III. PROPOSED SCHEME

The ADHD, dyslexia, dyscalculia, and dyspraxia prediction system based on ML and EAI is discussed in this part.

A. System Model

The system model for the ML based four neurodevelopment disorder predictions (dyslexia, ADHD, dyscalculia, and dyspraxia) is shown in Figure 1. Initially, the study uses a Google Form to gather information from instructors, parents, and patients on neurodevelopment disorders. Medical experts validate the information gathered. After then, this work used data preprocessing and feature engineering techniques. Following this, the data is split up into multiple sets for testing and training. The examination of ML models is conducted using testing data. The best ML scheme is chosen for the prediction of dyslexia, ADHD, dyscalculia, and dyspraxia based on the performance metrics results. The interpretability of ML models is then described using SHAP.

B. Dataset Preparation

A Google Forms questionnaire with a total of 48 questions was used to carry out the data collection phase. Patient's personal information and disease symptoms information were collected. 38 symptom-assessment items are included in the questionnaire to screen for dyslexia, ADHD, dyspraxia, and dyscalculia. The CLDQ model [17] and the ADHD rating scale in [16] were used as the foundation for the evaluation items. Three experts (professor and a medical doctor from Chittagong Medical College Hospital) validated the dataset and questionnaire. This study gathered 1,258 samples from a respondent group that included teachers, parents, and older siblings. According to the replies we gathered, the age distribution is rather evenly distributed throughout the four groups: 325 children (ages 6 to 10) and 325 (ages 11 to 15), 313 children (ages 3 to 5) and 295 teenagers (ages 16 to 18). Generally speaking, our dataset includes both numerical and category data.

Details of the questions used to obtain the dataset are shown in Figures 2(a), 2(b), and 2(c). A preset mapping was used to convert each child's responses to the symptom assessment items (such as never, rarely, sometimes, often, and always) into numerical scores based on the questionnaire data. The five domain-specific composites—social functioning, reading fluency, attention, hyperactivity, and arithmetic processing—were then created by combining these item-level scores. This paper then utilized threshold criteria to assign each composite score one of five diagnostic categories (dyslexia, ADHD, dyscalculia, dyspraxia, normal). According to the data gathered, 270 children were identified as having ADHD, 139 as having dyslexia, 146 as having dyspraxia, and 177 as having dyscalculia, while 526 children tested negative for any illness. This blatantly demonstrates the problems with class imbalance, which the SMOTE approach would address. A panel of three clinicians independently examined each labeled record during the dataset annotation process. The dataset was finally labeled using a majority vote. A preview of the dataset is shown in Figure 3(a). The dataset has a total of 53 columns and 1258 validated rows.

C. Data preprocessing

The data preprocessing stage consists of four main steps: exploratory data analysis, data transformation, data cleaning, and dataset balancing. The IQR approach was then used in this work to identify the outliers. In this phase, this paper used a box plot to examine whether any unusual or extreme values were present in the demographic and assessment answer columns. This work carefully managed missing and inconsistent data by removing incomplete entries and filtering out odd or conflicting responses in order to preserve the dataset's reliability. Data transformation involved converting raw questionnaire responses into standardized numeric values based on domain expertise and encoding categorical variables into numeric formats. All numerical features were normalized using Standard Scalar and Robust Scalar to have zero mean and unit variance in order to guarantee that each variable contributed equally during model training. Categorical data, such as age group, education level, and level, were converted into numerical format using label encoding or ordinal mapping. Label Encoder was used to transform the diagnostic category target variable into integer labels. Originally presented as qualitative ordinal categories (e.g., never, seldom, sometimes, often, and always), questionnaire responses were transformed into numerical scores using custom mapping methods. Where necessary, Standard Scalar was used to transform data distributions in order to enhance classifier performance across different feature types. The initial dataset showed a significant class imbalance, with the majority class accounting for 41.82% of samples and minority classes comprising between 11.02% and 21.48%. To handle the class imbalance issues, this paper used SMOTE approach (see Figure 3(b)).

D. Feature Engineering, Dataset splitting

Through systematic indexing, the 38 numerical assessment variables were divided into domain-specific subsets for the purpose of creating composite features. Row-wise mean aggregation across disorder-specific item clusters was used to calculate each composite feature. The feature aggregation method addresses feature redundancy by reducing multicollinearity. Based on accepted diagnostic criteria, disorder-specific weighting algorithms were used to create clinically-informed weighted features. Details of group-wise weighted composite features are captured in Figure 4(a). The construction details of expert-driven features are listed in Figure 4(b), and the weights allocated to each feature for are mentioned in Figure 4(c). To capture synergistic effects amongst disorder domains, interaction features were created by multiplying combinations of weighted and composite features. The key features were then identified using Recursive Feature Elimination with Cross-Validation (RFECV), where $k = 5$. The features importance score and the features used for training different machine learning models are displayed in Figures 4(d) and 4(e), respectively. Three distinct sets were created from our dataset. Seventy percent of data (880) went to training set, fifteen percent (189) is test set, and fifteen percent (189) is validation set.

Questions	Concised Feature Name		
Timestamp	timestamp	Often forgets daily routine tasks easily	adhd_q09_forgets_tasks
Email Address	email	Often twists arms, legs, or body while sitting	adhd_q10_fidgeting
Name of your child	child_name	Suddenly gets up from places where they are expected to sit	adhd_q11_leaves_seat
Age	age	Often runs around or climbs on things	adhd_q12_run_climbs
Education Level	education_level	Finds it difficult to play calmly	adhd_q13_quiet_play
Address	address	Always seems to be on the go or as if driven by a motor	adhd_q14_always_moving
Division	division	Talks more than necessary	adhd_q15_excessive_talking
Does your child go to school?	goes_to_school	Answers questions before they are fully asked	adhd_q16_blurts_answers
Institution	institution	Finds it difficult to wait for their turn	adhd_q17_waiting_turn
Class	child_class	Disrupts or intrudes on others (e.g., interrupts in	adhd_q18_intrudes_others
Your child often fails to focus on simple things or makes mistakes negligently	adhd_q01_focus_mistakes	Difficulty in spelling?	dyslexia_q01_spelling_difficulty
Often struggles to hold attention on daily or sport activities	adhd_q02_attention_activities	Difficulty in learning letters?	dyslexia_q02_letter_learning
While you are talking to him/her, it seems like he/she is not listening	adhd_q03_listening_attention	Difficulty in learning phonics (sounding out words)?	dyslexia_q03_phonics_difficulty
Does not follow instructions properly and is not able to finish tasks	adhd_q04_instruction_following	Reads slowly?	dyslexia_q04_reading_speed
Faces problems or finds it inconvenient to work in an organized way	adhd_q05_organization_difficulty	Reads below grade or expectancy level?	dyslexia_q05_reading_level
Often ignores or strongly dislikes activities requiring sustained attention	adhd_q06_sustained_attention	Required extra help in school for reading or spelling?	dyslexia_q06_extra_help
Loses necessary items such as pencil, book, toy, etc.	adhd_q07_loses_items	Handwriting is spatially disorganized?	dyslexia_q07_handwriting_spatial
Often gets distracted by outside stimuli	adhd_q08_external_distraction	Papers look disorganized or messy?	dyslexia_q08_paper_organization
		Drawings look immature for age?	dyslexia_q09_drawing_maturity
		Poor understanding of interpersonal space?	dyspraxia_q01_interpersonal_space
		Difficulty knowing how others react?	dyspraxia_q02_social_reactions
		Trouble understanding others' feelings?	dyspraxia_q03_understand_feelings

(a)

(b)

Fig. 2. Questions used (dataset collection purpose)

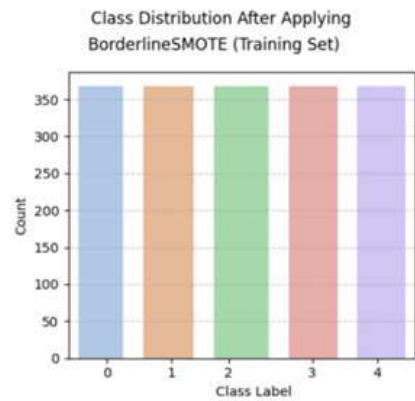
age_group	education_level	Division	adhd_q01_focus_mistakes	adhd_q02_attention_activities	adhd_q03_listening_attention
6-10 years old	Primary	Chattogram	Sometimes (0)	Rarely or not at all (0)	Sometimes (0)
3-5 years old	Primary	Chattogram	Rarely or not at all (0)	Always or frequently (1)	Sometimes (0)
11-15 years old	Secondary	Chattogram	Sometimes (0)	Sometimes (0)	Rarely or not at all (0)
3-5 years old	Primary	Dhaka	Sometimes (0)	Sometimes (0)	Rarely or not at all (0)
6-10 years old	Primary	Chattogram	Sometimes (0)	Rarely or not at all (0)	Rarely or not at all (0)
6-10 years old	Primary	Dhaka	Sometimes (0)	Sometimes (0)	Rarely or not at all (0)
3-5 years old	Primary	Dhaka	Rarely or not at all (0)	Sometimes (0)	Sometimes (0)
3-5 years old	Primary	Chattogram	Sometimes (0)	Rarely or not at all (0)	Sometimes (0)
3-5 years old	Primary	Dhaka	Always or frequently (1)	Often (1)	Rarely or not at all (0)
11-15 years old	Secondary	Chattogram	Often (1)	Rarely or not at all (0)	Rarely or not at all (0)
11-15 years old	Primary	Dhaka	Rarely or not at all (0)	Sometimes (0)	Rarely or not at all (0)
3-5 years old	Primary	Chattogram	Sometimes (0)	Always or frequently (1)	Rarely or not at all (0)
3-5 years old	Primary	Chattogram	Rarely or not at all (0)	Rarely or not at all (0)	Always or frequently (1)
3-5 years old	Primary	Chattogram	Often (1)	Rarely or not at all (0)	Rarely or not at all (0)
16-18 years old	Higher Secondary	Chattogram	Sometimes (0)	Rarely or not at all (0)	Often (1)

(a)

Fig. 3. (a) Dataset portion, (b) SMOTE apply for class distribution

Questions	Concised Feature Name
Makes inappropriate comments showing lack of social understanding?	dyspraxia_q04_social_comments
Difficulty making or keeping friends?	dyspraxia_q06_friendship_difficulty
Isolates self in social situations?	dyspraxia_q06_social_isolation
Fees anxious or out of place in new social situations?	dyspraxia_q07_social_anxiety
On arithmetic problems, has difficulty keeping numbers in columns?	dyscalculia_q01_number_columns
Worse at math than reading and spelling?	dyscalculia_q02_math_vs_reading
Makes careless math errors (e.g., adding instead of subtracting)?	dyscalculia_q03_calculation_errors
Trouble learning new math concepts?	dyscalculia_q04_concept_learning

(c)



(b)

New Created Features	Feature's Name	Weight Assigned
dyslexia_weighted	dyslexia_q04_reading_speed	0.40
	dyslexia_q03_phonics_difficulty	0.35
	dyslexia_q01_spelling_difficulty	0.25
dyspraxia_weighted	dyspraxia_q06_social_isolation	0.40
	dyspraxia_q07_social_anxiety	0.35
dyscalculia_weighted	dyscalculia_q02_math_vs_reading	0.60
	dyscalculia_q01_number_columns	0.40

(a)

Expert Driven Feature Construction Details

Created Feature Name	Feature Details
ADHD_weighted	age_numeric, adhd_q02_attention_activities_numeric, adhd_q09_forgets_tasks_numeric, adhd_q14_always_moving_numeric

(b)

Assigned weights per features

Feature/Age Group	3-5 year	6-10 year	11-15 year	16-18 year
adhd_q02_attention_activities_numeric	0.25	0.33	0.40	0.45
adhd_q09_forgets_tasks_numeric	0.25	0.33	0.40	0.45
adhd_q14_always_moving_numeric	0.50	0.34	0.20	0.10

(c)

ADHD_composite	0.096130	Number of features: 16
Dyslexia_composite	0.064010	Feature list: ['dyslexia_q03_phonics_difficulty_numeric',
Dyscalculia_composite	0.044515	'dyslexia_q04_reading_speed_numeric',
Dyspraxia_composite	0.039107	'dyslexia_q01_spelling_difficulty_numeric',
Dyspraxia_vs_Math	0.025068	'dyspraxia_q06_social_isolation_numeric',
dyscalculia_weighted	0.014047	'dyscalculia_q02_math_vs_reading_numeric', 'ADHD_composite',
adhd_dyslexia_interaction	0.009364	'dyslexia_composite', 'Dyspraxia_composite',
dyspraxia_weighted	0.008162	'Dyscalculia_composite', 'Dyspraxia_vs_Math',
adhd_weighted	0.007630	'adhd_dyslexia_interaction', 'dyspraxia_age_interaction',
dyslexia_q03_phonics_difficulty_numeric	0.007386	'adhd_weighted', 'dyslexia_weighted', 'dyspraxia_weighted',
dyslexia_q01_spelling_difficulty_numeric	0.006847	'dyscalculia_weighted']
dyslexia_q04_reading_speed_numeric	0.006598	
dyscalculia_q02_math_vs_reading_numeric	0.006413	
dyspraxia_q06_social_isolation_numeric	0.005421	
dyspraxia_age_interaction	0.005387	

(d) Feature importance score of the features

(e) Selected Features via RFECV

Fig. 4. (a) Group wise feature weighting, (b) expert driven feature, (c) assigned weight, (d) feature importance, (e) selected feature via RFE

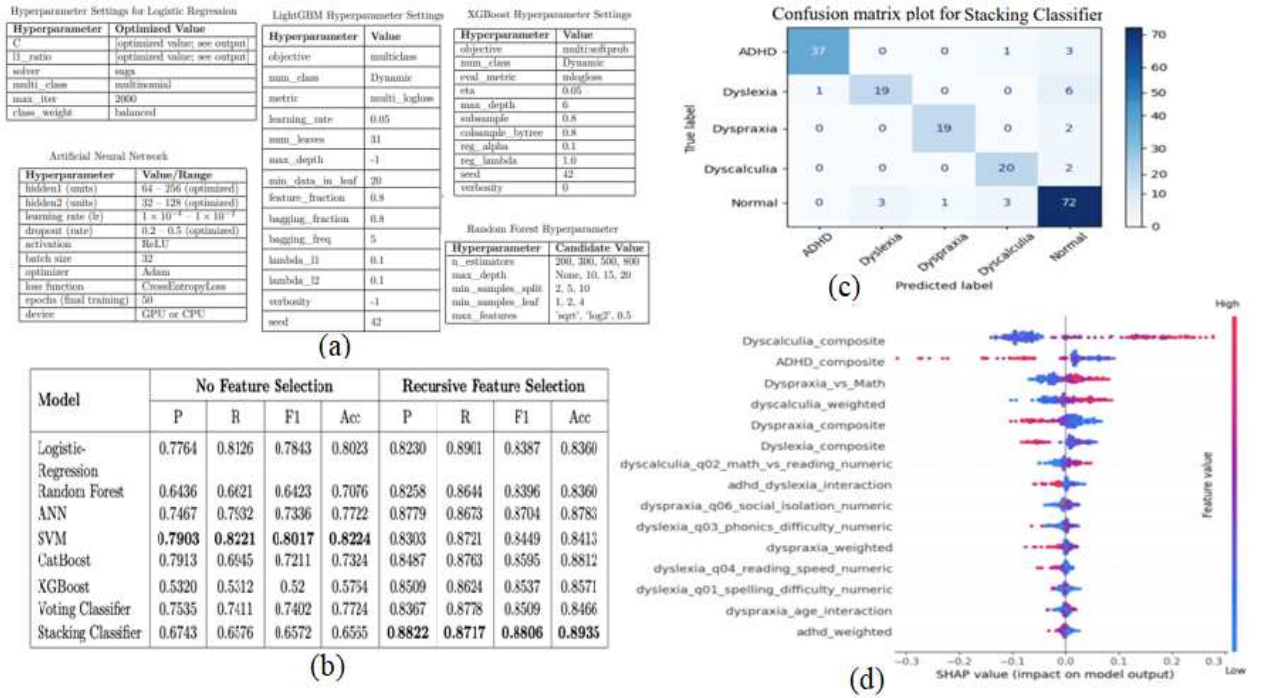


Fig. 5. (a) Hyper parameter setting, (b) ML model selection, (c) Stacking classifier confusion matrix, (d) SHAP result

TABLE I
COMPARISON RESULT

Work	Accuracy value	Precision	Recall value	F1 value
X. Zhang et al., [7]	85.29%	86.09%	85.39%	85.71%
Bhushan et al. [9]	72.99%	70.27%	70.81%	73.02%
Zaibi et al.,[12]	79.91%	75.59%	77.43%	79.89%
Midhun et al.,[8]	87.30%	86.17%	86.49%	87.83%
Proposed Method	89.35%	88.22%	87.17%	88.06%

D. Model selection, Hyper parameter tuning

This study employed a number of hyper parameter tuning techniques to get the best results (e.g., random search and grid search). Figure 5(a) displays the hyper parameter parameters for various machine learning models. This study evaluated two ensemble ML models (voting and stacking classifier) and six individual ML models (ANN, logistic regression or LR, SVM, CatBoost, random forest or RF, and XGBoost) in order to determine the optimum ML model selection for neurodevelopment disease prediction. The LightGBM, XGBoost, and CatBoost models are part of the voting classifier. RF, LightGBM, XGBoost, ANN, and LR models are all included in the stacking ensemble approach.

A comparison of ML model performance with and without feature selection is presented in Figure 5(b). The stacking classifier outperforms other ML models in all compared performance matrices (e.g., accuracy) when using recursive feature selection. Figure 5(c) displays the confusion matrix. It is evident from both figures that the stacking ensemble approach provides superior AUC values for every prediction class together with a greater true positive rate. A SHAP plot, shown in Figure 5(d), illustrates the feature impact on the prediction output. Each point displays the Shapley value for that feature for a single validation sample, indicating the feature's amplitude and direction of effect. The figure shows that the Dyscalculia composite feature was given the highest weight by the model when forecasting output.

IV. RESULTS AND ANALYSIS

Table I outlines that the proposed stacking ensemble technique achieves accuracy value of 89.35% and F1 value of 88.06%, outperforming other compared studies (e.g., [7], [8], [9], [12]). The accuracy and precision value of the suggested work are more than 2% greater than those of the current work. The proposed stacking ensemble approach outperforms other approaches due to its sophisticated dataset preparation, SMOTE-based dataset imbalance problem solutions, RFE-based feature selection, hyper parameter tweaking, and grid and random search approaches.

V. CONCLUSION

A ML based dyslexia, ADHD, dyscalculia, and dyspraxia prediction system is presented in this paper. This paper used Borderline-SMOTE to balance minority classes and created a domain-informed feature engineering framework with age-stratified, expert-weighted composite scores and interaction terms. The important feature selection process is completed by using RFE approach and model performance was maximized by using hyper parameter tweaking (using grid search, randomized search). With 89.35% accuracy, 88.22% precision, 87.17% recall, and an 88.06% F1-score, the findings demonstrated that the suggested stacking ensemble technique, which combines Random Forest, XGBoost, LightGBM, ANN, and logistic regression, produced the best results. The results hinted that the proposed stacking ensemble methods performs well than previous studies, attaining accuracy and precision values of at least 2%. Furthermore, SHAP-based explain ability improved our predictive system's clinical interpretability and dependability by offering clear insights into feature contributions. Future research challenges include, but are not limited to, FL, quantum computing, LLM-based medication recommendations for various neurodevelopment disorders, block chain-based data privacy and security enhancement for patients with neurodevelopment disorders, and IoT and DL-based real-time neurodevelopment disorder detection.

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